

Artificial Intelligence & Generative AI

Assoc. Professor Darwin Kim Bulawan



Course Map

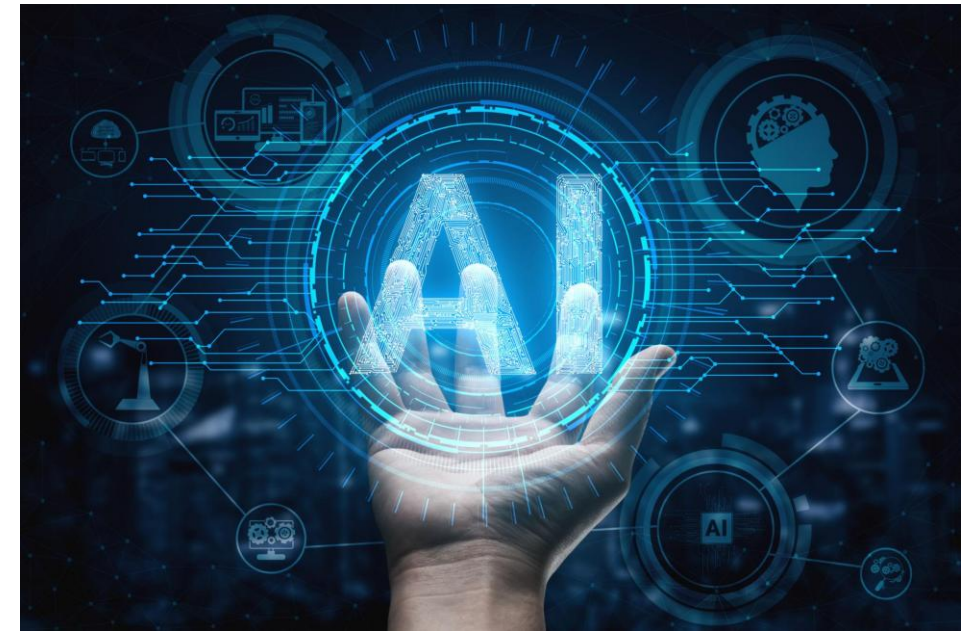
Week Number	Inclusive Dates	Module Number	Assessments	Topics
1		1	-	Course Introduction, Discussion of Deliverables, Framework for Analyzing IT Trends, Global IT Landscape
2-3		2	Analysis Paper (I)	Artificial Intelligence (AI) & Generative AI, Large Language Models, Enterprise AI Adoption, AI Ethics, Bias and Explainability, AI Regulation (EU AI Act, PH AI initiatives)
4-5		3	Presentation (G)	Cybersecurity Trends, Zero Trust Architecture, Ransomware Evolution, Cloud-Native Architecture, Multi-Cloud Strategy, Edge Computing
6-7		4	Review Paper (I)	Data Governance, Data Privacy (Philippine DPA updates), Data Ethics & Responsible Analytics, Blockchain & Web3, Smart Contracts, Enterprise Adoption vs Speculation
8		5	Plan Report (G)	Internet of Things (IoT), Smart Systems & Industry 4.0, IoT Security Challenges
9		-	Exam	MIDTERM EXAMINATION
10-11		6	Research Paper (I)	Automation & Hyperautomation, RPA & Process Mining, Low-Code / No-Code Platforms, Quantum Computing, Post-Quantum Cryptography
12-13		7	Proposal Paper (G)	Sustainable IT, Green Computing, ESG & Digital Responsibility, DevOps & DevSecOps, Platform Engineering, GitOps
14-15		8	Analysis Report (I)	Extended Reality (AR/VR/MR), Spatial Computing, Human-Computer Interaction Trends, Digital Transformation in the Philippines, eGovernment, FinTech & Digital Banking
16-17		9	Presentation (G)	IT Project Failures, Governance Breakdowns, Ethical and Strategic Risks
18		-	Exam	FINALEXAMINATION



Learning Objectives

In the end of this session, students will be able to:

- Explain what Artificial Intelligence and Generative AI are
- Describe how Large Language Models work at a high level
- Analyze enterprise adoption of AI
- Identify risks and ethical concerns
- Evaluate AI's long-term impact on IT careers

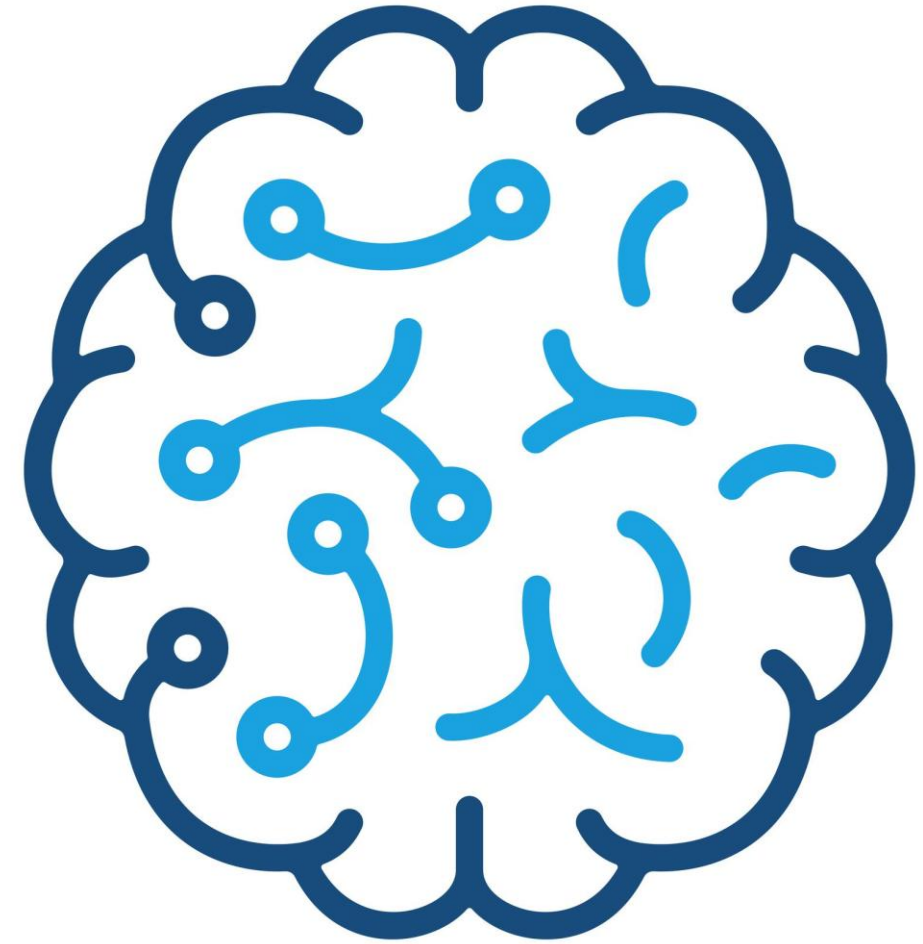


What is Artificial Intelligence?

AI is the simulation of human intelligence by machines.

Artificial Intelligence:

- Machines performing tasks requiring human intelligence
- Includes learning, reasoning, and decision-making
- Uses data to improve performance over time



Narrow AI vs General AI

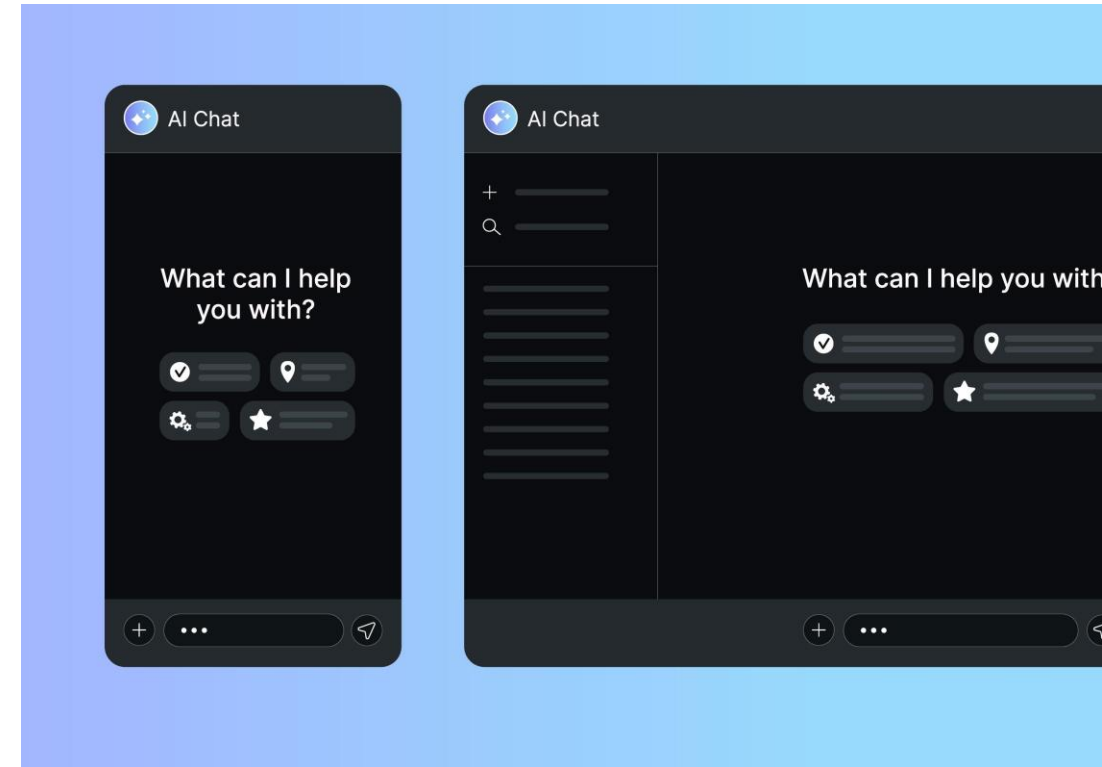
Not all AI is equal. Most AI today is narrow and specialized.

Narrow AI:

- Designed for specific tasks
- Examples: recommendation engines, chatbots

Artificial General Intelligence (AGI):

- Hypothetical AI with human-level reasoning
- Not yet achieved

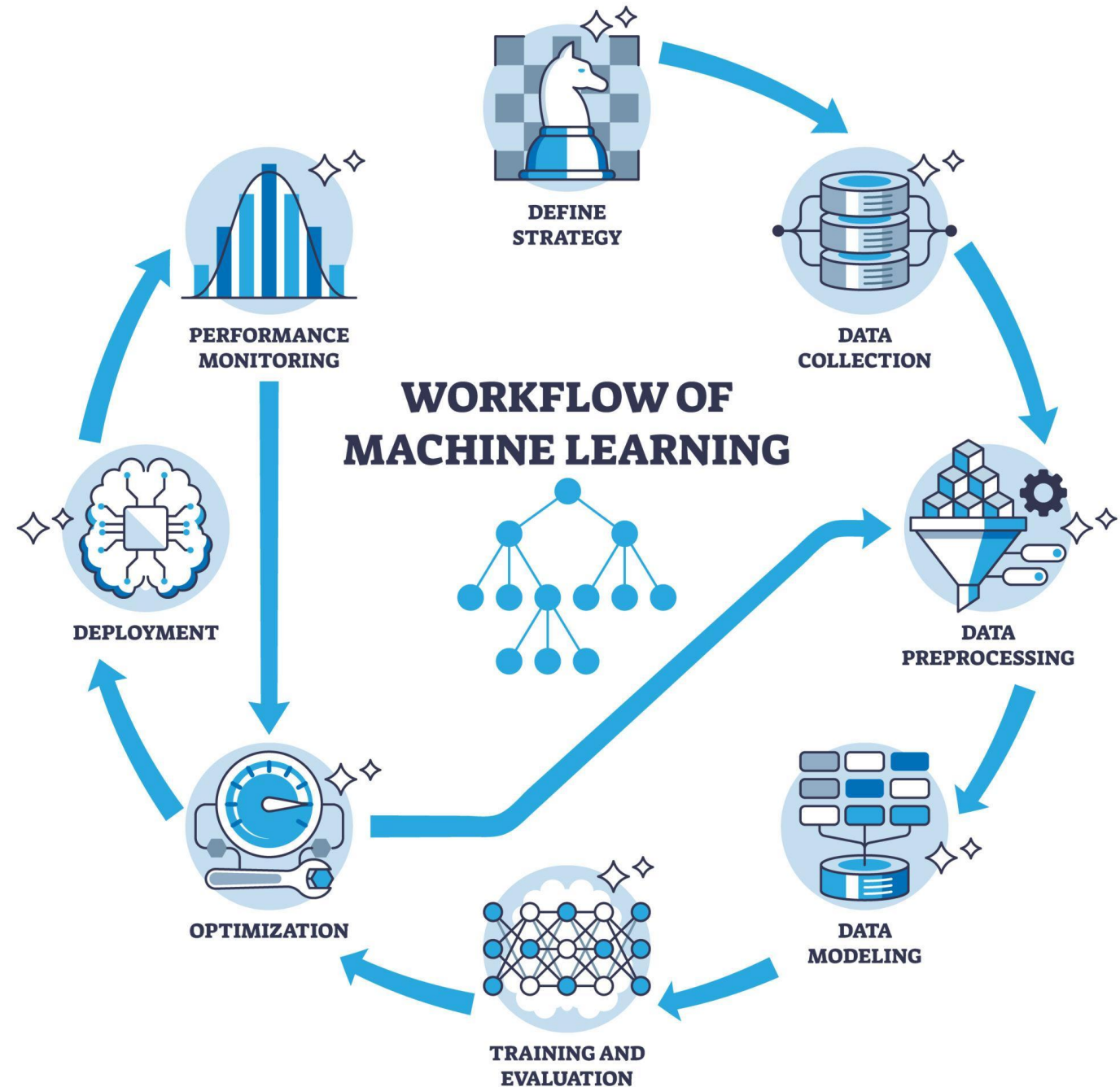


Machine Learning Basics

Machine Learning is the engine behind modern AI.

Machine Learning:

- Systems learn patterns from data
- Improves without explicit programming
- Types: supervised, unsupervised, reinforcement

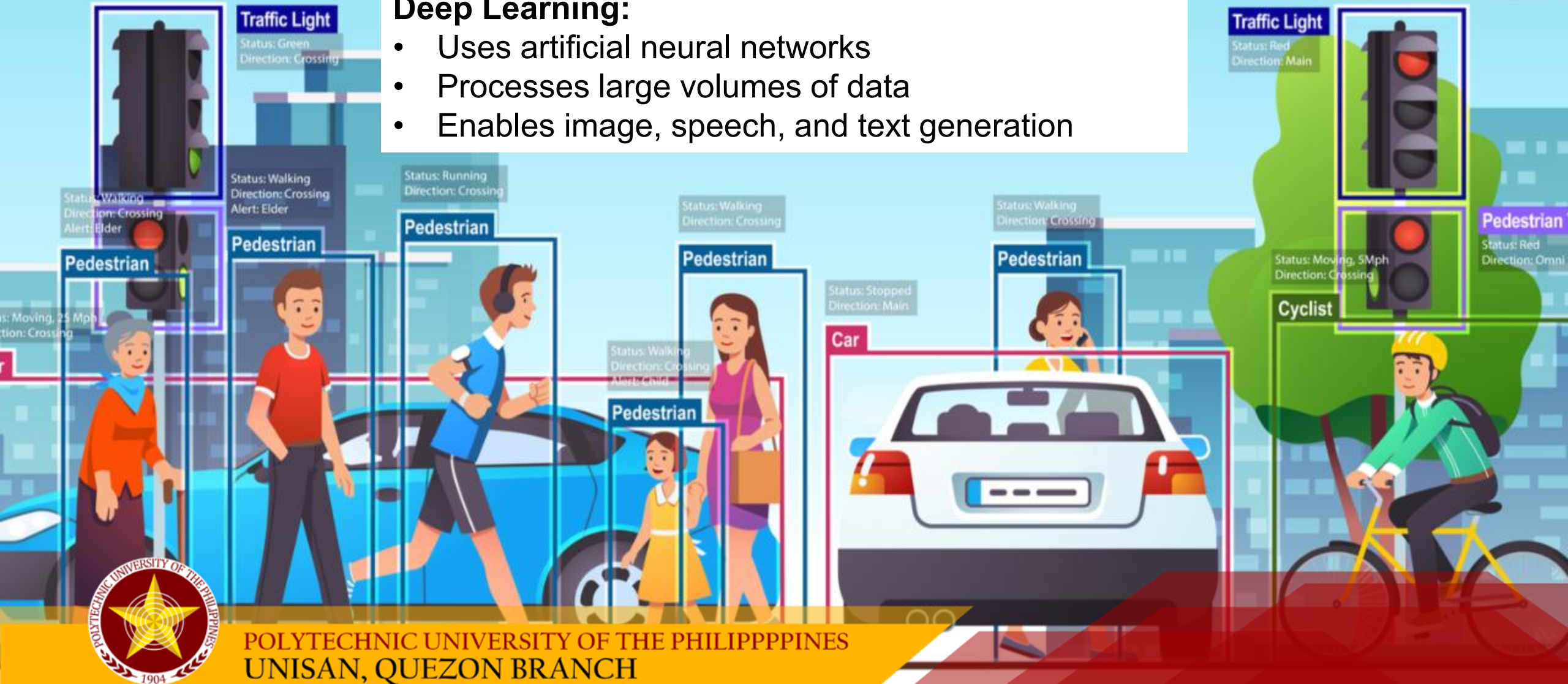


Deep Learning

Deep Learning powers most generative AI systems.

Deep Learning:

- Uses artificial neural networks
- Processes large volumes of data
- Enables image, speech, and text generation

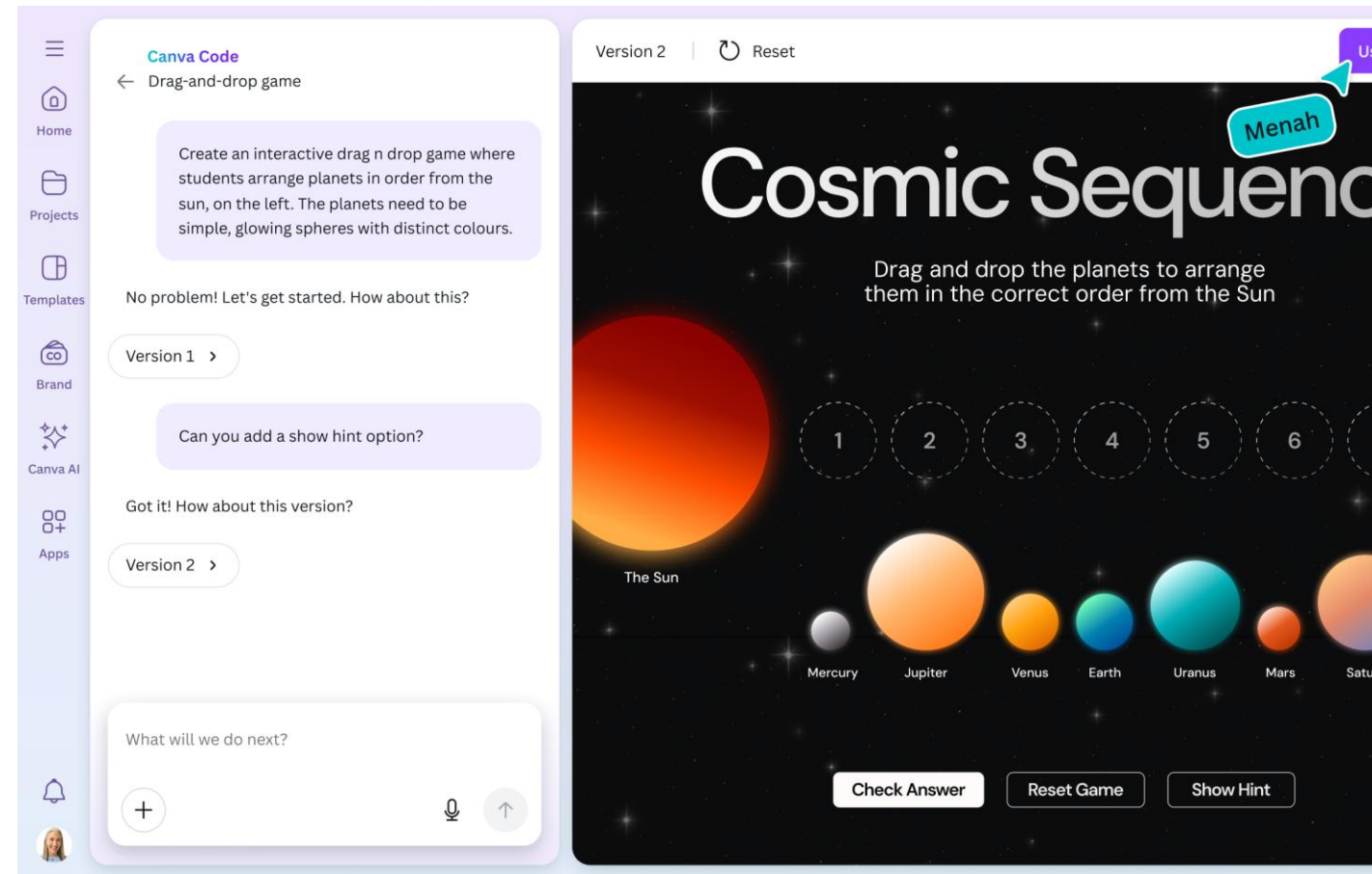


What is Generative AI?

Generative AI creates new content instead of just analyzing data.

Generative AI:

- Produces text, images, code, audio
- Learns from massive datasets
- Predicts next element in sequence

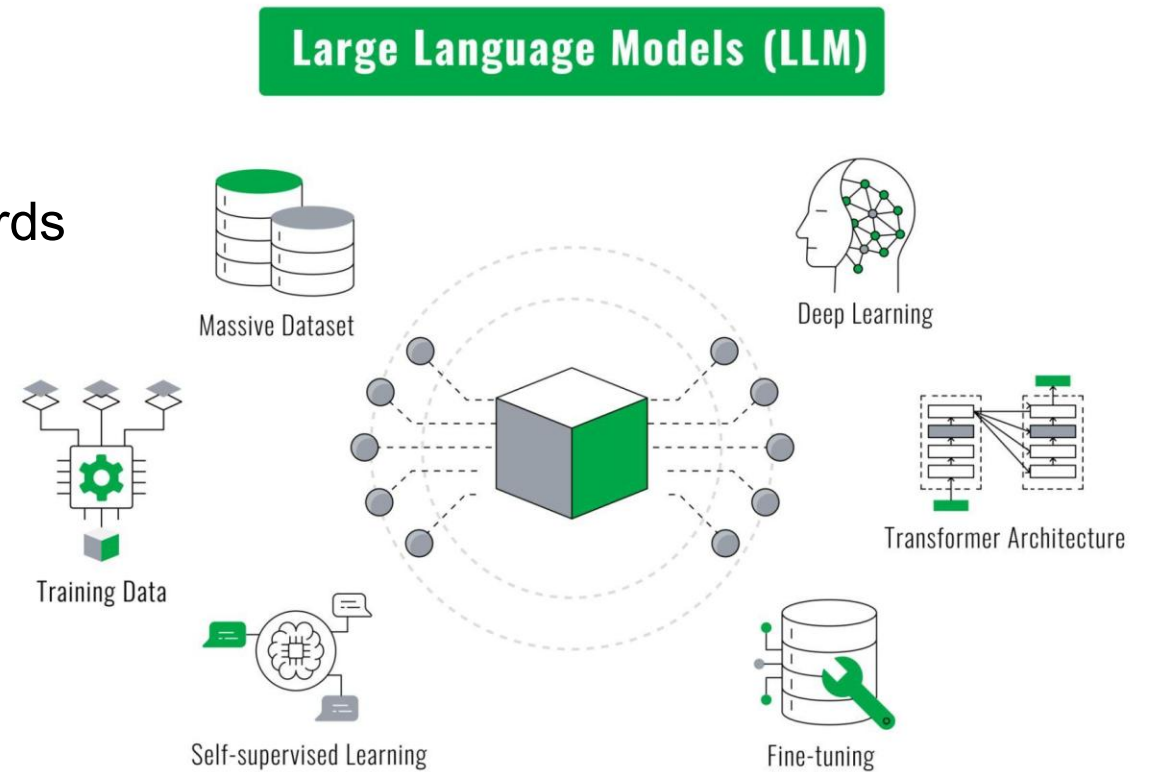


Large Language Models (LLMs)

LLMs are foundation models trained on massive text data.

LLM:

- Neural network trained on billions of words
- Learns grammar, context, patterns
- Used in ChatGPT, Gemini, Claude



Enterprise Adoption of AI

AI is now embedded in business workflows.

Enterprise AI:

- AI copilots for productivity
- Customer service automation
- Predictive analytics
- AI-driven decision systems



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AI in Software Development

AI is transforming coding workflows.

AI in Dev:

- Code completion
- Bug detection
- Automated testing
- Documentation generation

Top 8 **Generative AI Tools** For Software Development In 2024

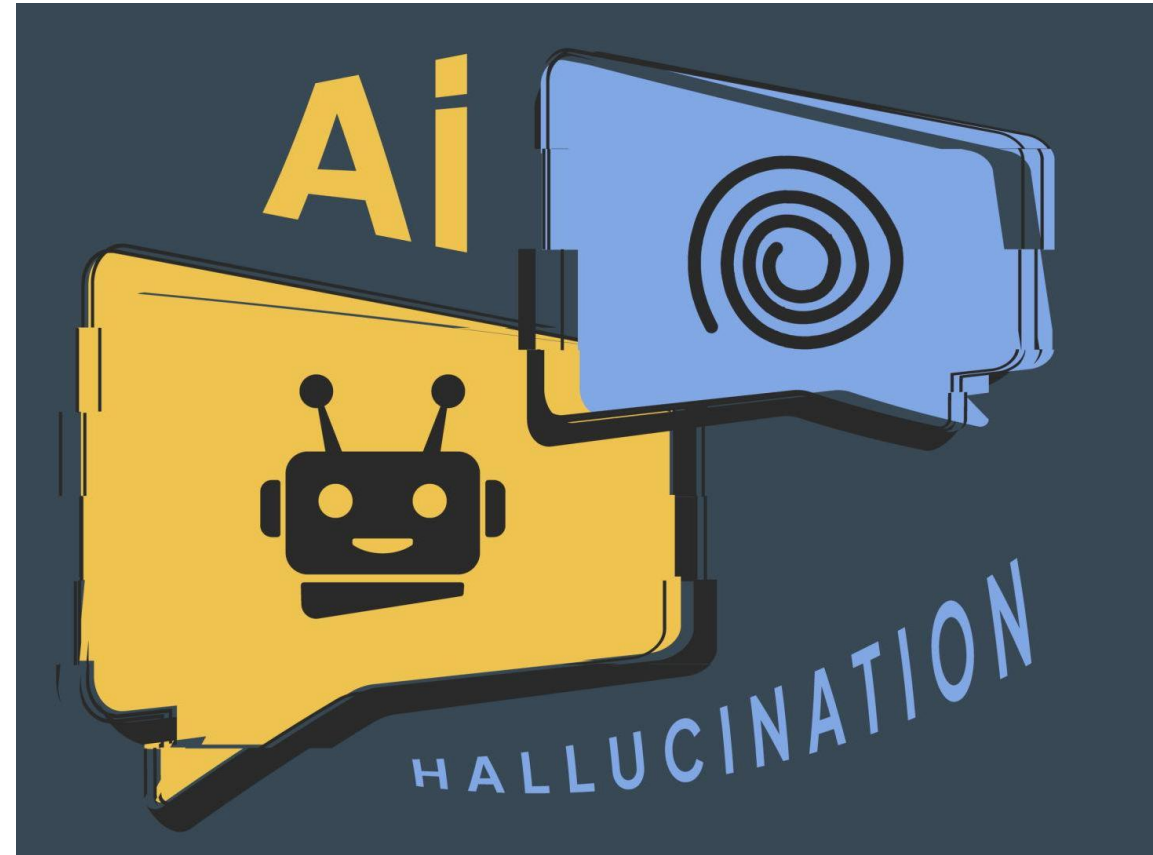


AI Risks: Hallucination

AI can confidently produce incorrect information.

Hallucination:

- Fabricated but plausible outputs
- Occurs due to probabilistic prediction
- Risk in legal, medical, academic use

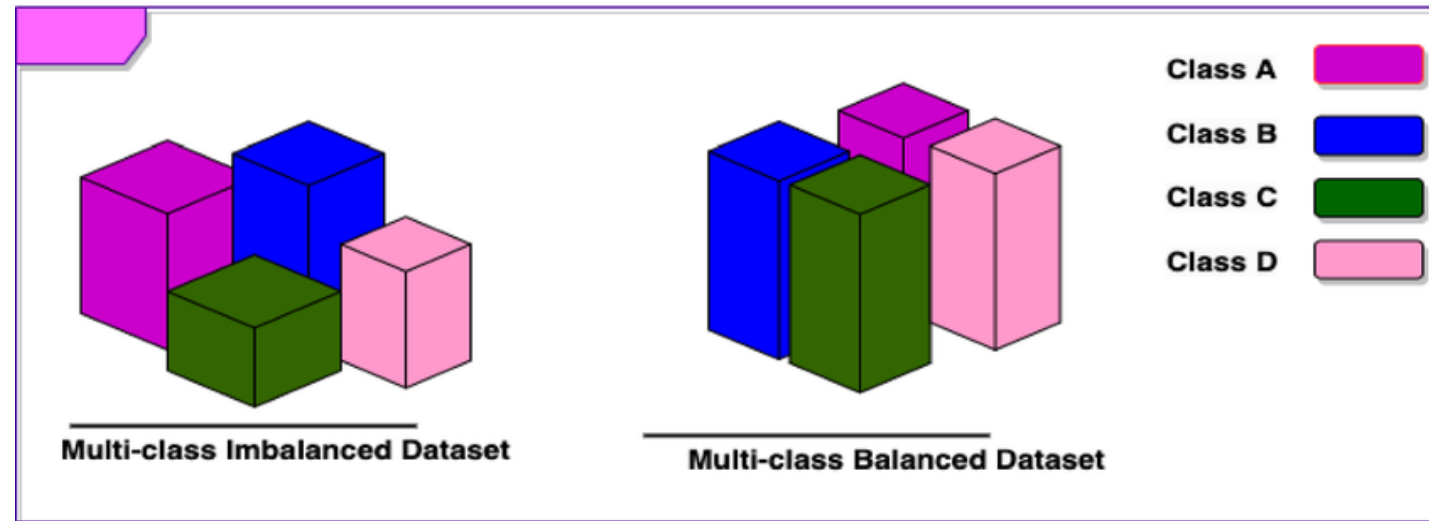
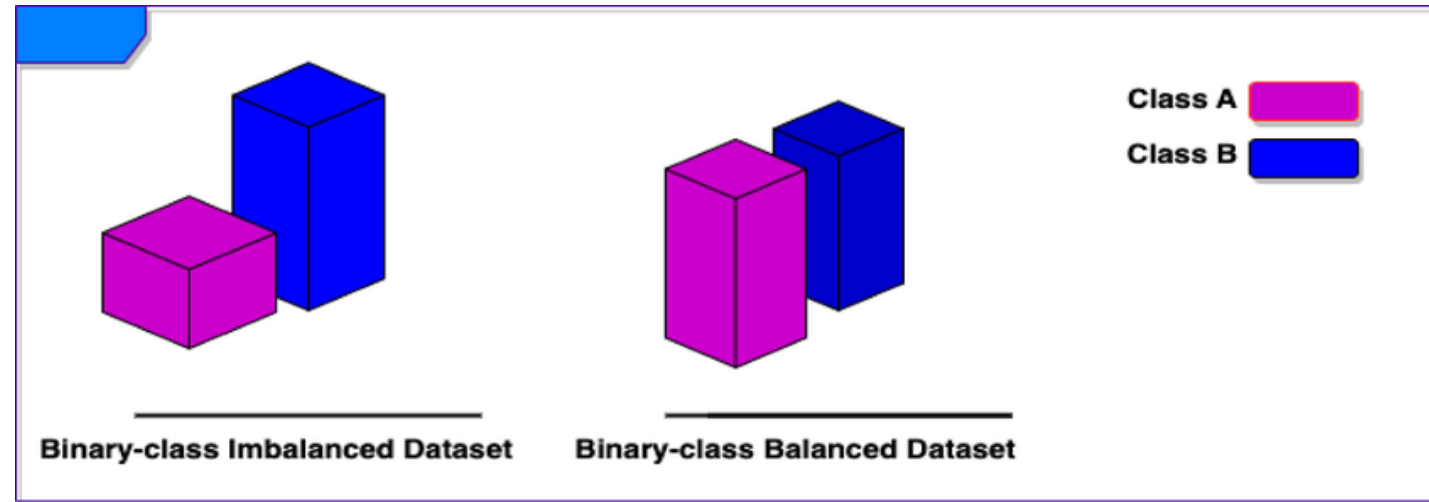


AI Bias

AI reflects biases present in training data.

AI Bias:

- Discriminatory outputs
- Skewed predictions
- Unequal representation in datasets

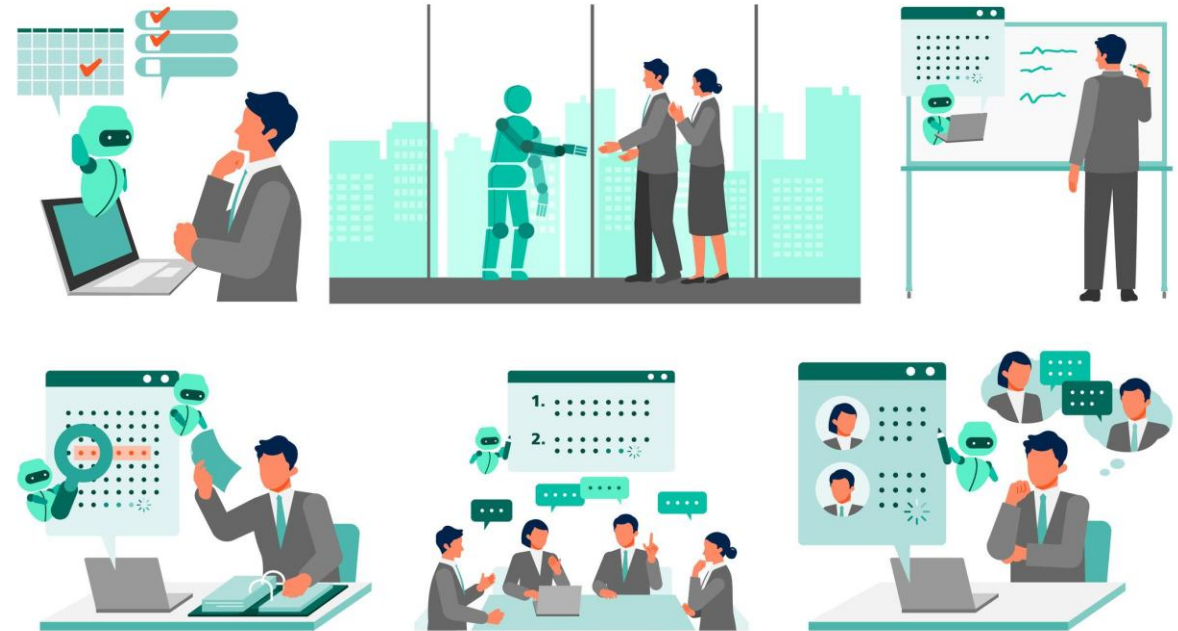


AI and Job Displacement

AI reshapes workforce roles.

Workforce Impact:

- Automates repetitive tasks
- Creates AI supervision roles
- Increases demand for AI literacy



AI Regulation

Governments are responding to AI growth.

AI Regulation:

- EU AI Act
- Risk-based classification
- Compliance requirements



AI in the Philippines

AI adoption is increasing locally.

PH Context:

- AI in BPO
- AI in fintech
- DICT AI roadmap



AI Governance Frameworks

As AI systems grow more powerful, organizations must implement structured governance to manage risk and accountability.

Governance ensures AI is used responsibly, ethically, and strategically.

AI Governance:

- Structured policies guiding AI use
- Risk management processes
- Accountability mechanisms
- Transparency standards



Responsible AI Principles

Responsible AI ensures fairness, transparency, and safety.

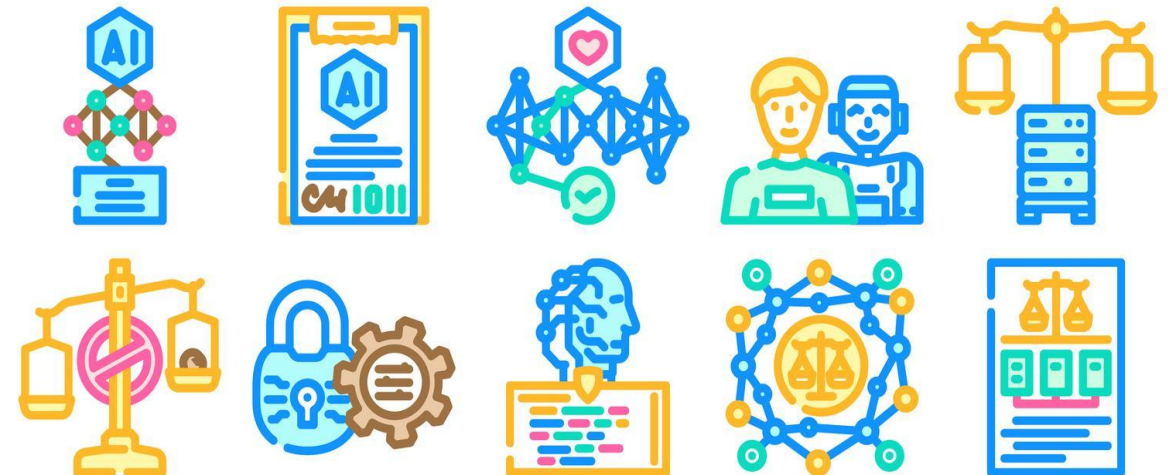
It focuses on building trust in AI systems.

Responsible AI:

- Fairness (no discrimination)
- Transparency (explainable outputs)
- Accountability (clear ownership)
- Safety and reliability



ETHICAL AI



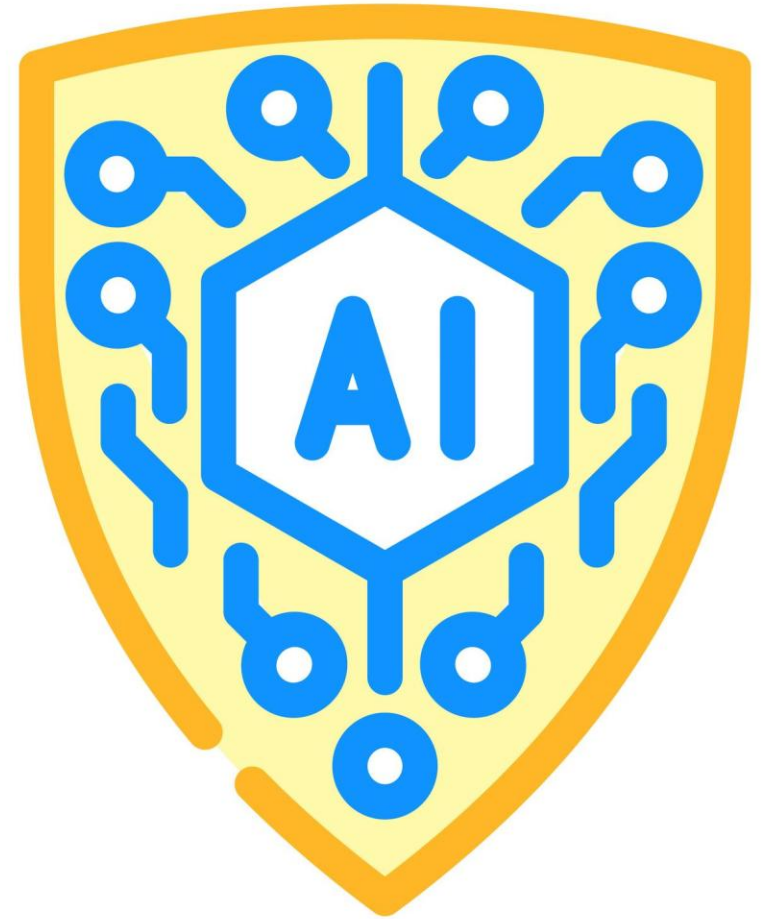
AI Security Risks

AI systems themselves can become targets of attack.

Security must evolve with AI capabilities.

AI Security Risks:

- Data poisoning
- Model theft
- Prompt injection
- Adversarial attacks



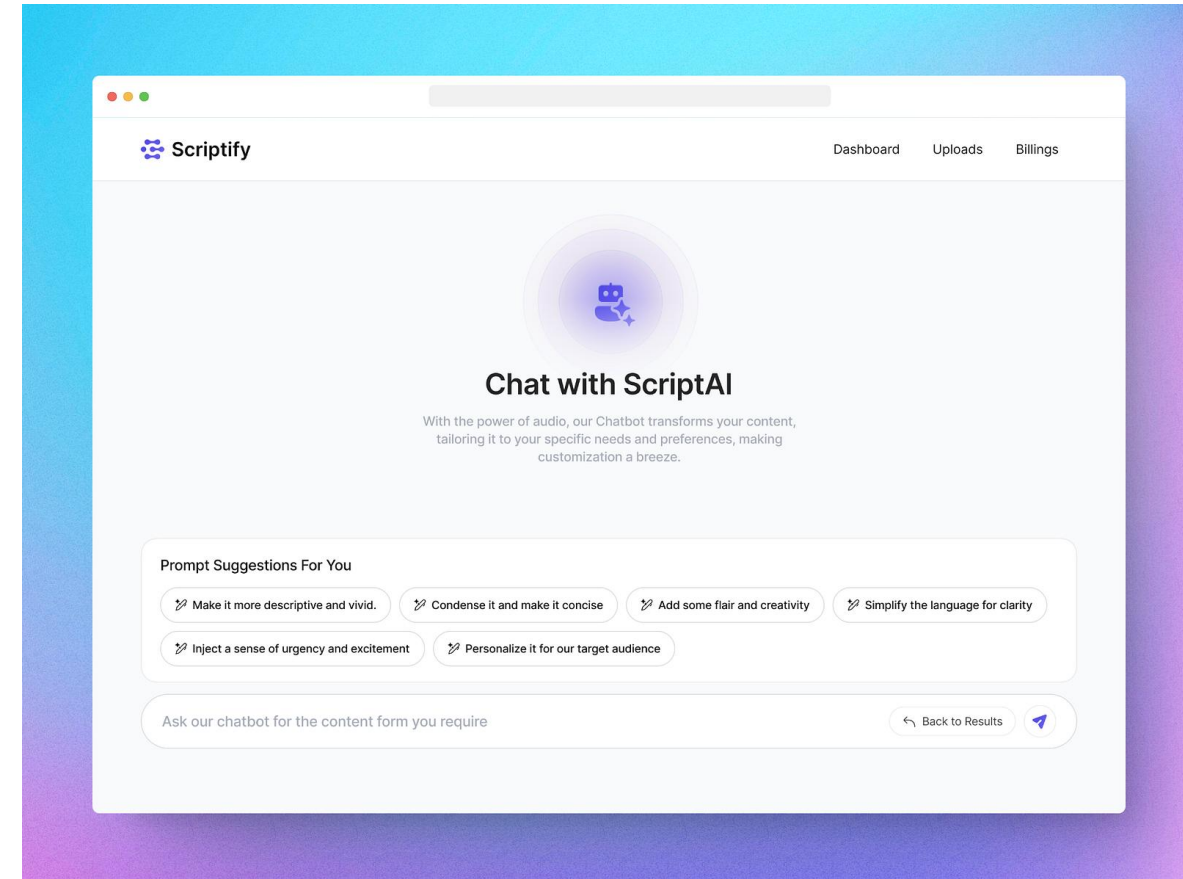
Prompt Engineering Basics

The quality of AI output depends on the quality of the prompt.

Prompting is becoming a key professional skill.

Prompt Engineering:

- Structured instructions to AI
- Clear context improves accuracy
- Iterative refinement process



Limitations of AI

AI is powerful but not omniscient. Understanding limitations prevents misuse.

AI Limitations:

- No real understanding
- Dependent on training data
- Prone to hallucination
- Context limitations

✦✦ Explore this report with AI BETA



Hello! I'm a virtual agent here to help you explore the key findings of the IEA's special report on energy and AI. Remember, AI can make mistakes, and you should verify the answers independently with the original source materials.

Just now

Type your message

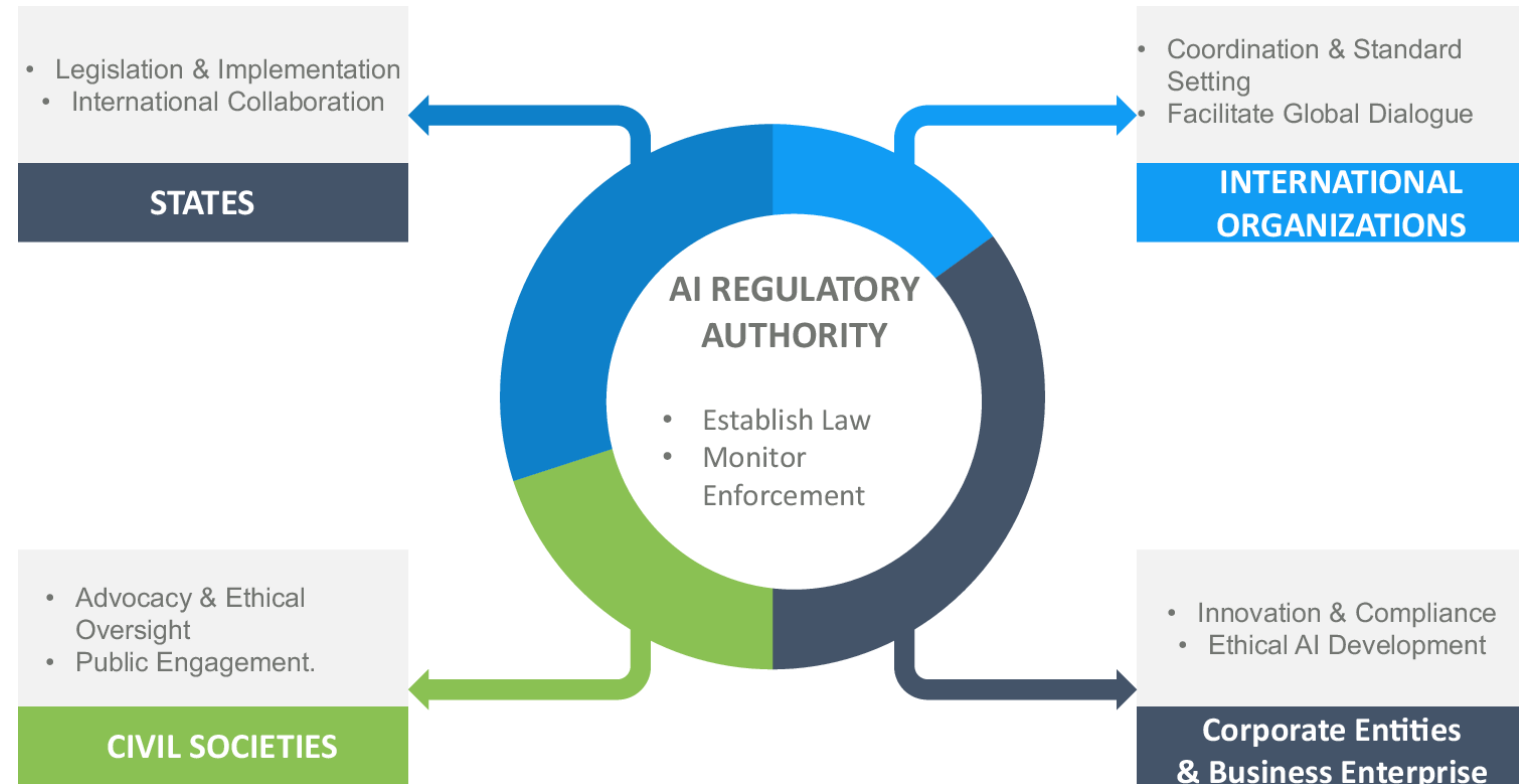


Strategic Outlook: AI in 2026

AI integration will deepen across industries. Future readiness requires adaptability.

Strategic Outlook:

- AI copilots everywhere
- Increased AI regulation
- Hybrid human-AI teams
- AI-specialized job roles



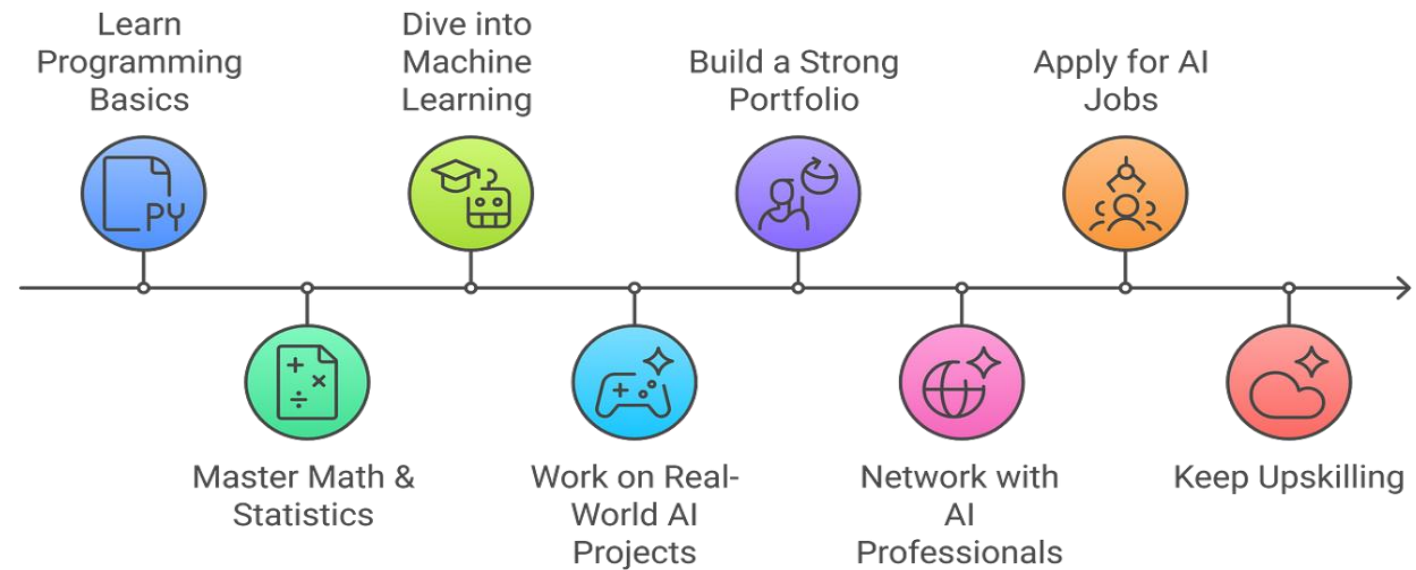
AI and Lifelong Learning

The AI era requires continuous skill upgrading.
Static knowledge becomes outdated quickly.

Lifelong Learning:

- Continuous upskilling
- AI literacy
- Ethical awareness
- Cross-disciplinary knowledge

Path to Becoming an AI Engineer



Final Synthesis

AI is transformative, but complex.

As IT students, your role is not to fear it or worship it, but to understand it.

Key Takeaways:

- AI is data-driven pattern recognition
- Generative AI predicts and creates
- Enterprise adoption is accelerating
- Risks and governance matter
- Critical thinking defines IT professionals



Module 2 Activity (Individual)

Trend Analysis Brief (AI Focus)

What LLM did you use to accomplish this activity? (i.e., ChatGPT, Gemini, etc.)

What prompt did you use to complete this activity?

Guidelines/Requirements:

1. Upload your activity in .pdf
2. Use official PUP document template
3. Generate your activity covering:

Analyze AI trend impact

500-700 words

Include business + ethical implications

Cite at least 3 credible reports

